

Grok Star Lab

Baseline vs Upgrades

A side-by-side comparison of the original control plane
against each upgrade shipped on this machine

Repo: ~/Projects/grok-home · **Branch:** main

Date: 2026-08-01 · **Tip:** 9e62600 (embedding drift)

Doctor: OPERATIONAL

How to read this document: each section takes one capability as it existed in the **baseline** Star Lab / Grok home setup, then shows the **upgraded** form after the recent build program (token policy, SQC, research factory, graph routing, progressive delivery, mind/body design, and model packaging standards).

1. Executive summary

Baseline Grok Star Lab already provided a free, Mac-local control plane: **lab tokens** (EV routing with λ -kilotokens), forge/gym/showroom modules, SessionStart token boot, and doctor health checks. What it lacked was closed-loop proof, multi-agent context discipline, progressive delivery gates, a product doctrine for defensible agents (mind vs body), and a portable model/tokenizer package contract.

The upgrades do not replace Grok as the frontier mind. They make **when** and **how** the mind is used legible, measurable, and reversible—and they define what persists when the model is swapped.

1.1 Stack at a glance

Area	Baseline (before)	Upgrade (now)
North-star goal	Ship features; tokens are an afterthought	Cost per successful end-to-end task; observation before influence
Routing	Heuristic EV modes (local/short/medium/deep)	Same EV core + shadow dual-log + frozen golden + four-stage rollout
Quality labels	No hard gate on distill	SQC Loop 3 must accept before lab tokens distill
Multi-agent context	Mode packing knobs only (max_context, subagents)	RCR-style lab graph: role budgets, 3-gate score, pass/summarize/drop
Failure handling	Hope tests pass; no standard recovery	LEAD recover + forge --recover + circuit breakers
Product doctrine	Lab as toolbelt / chatbot-wrapper risk	Mind + body: identity, property, contact, limits
Local model artifacts	Ad hoc .pt / pickle risk	safetensors + manifest + tokenizer-major + drift gates

2. Baseline snapshot

For this report, **baseline** means the integrated Star Lab on main after the initial token-aware control plane and module matrix—before the closed-loop research program, graph L2 routing, progressive delivery, mind/body designs, and weights packaging stack.

- lab tokens route / complete / audit / distill (heuristic LenVM-style horizon)
- SessionStart hook arming GROK_TOKEN_POLICY and default short posture
- forge, gym (dolphin3), showroom, knowledge FTS, arena, ship, design CLI
- doctor pass matrix for core modules; local portal when GitHub unavailable
- EV formula: reward – λ ·kilotokens; packing plan per mode

3. Upgrade-by-upgrade comparison

Each subsection pairs the baseline behavior with the upgraded system. Commits are tips on main.

3.1 Annotation quality — SQC (Loop 3)

Primary commit: b3b6f26

Area	Baseline (before)	Upgrade (now)
Quality control	No formal annotation acceptance sampling	lab sqc risk sample loop asn — single/double/sequential plans, risk scoring
Distill gate	tokens distill could run on any audits	Hard SQC gate: distill blocked unless last loop quality_sufficient (or --ungated)

Area	Baseline (before)	Upgrade (now)
Operator UX	Manual judgment only	Loop log + diagram; KPI tracks sqc_pass_rate / last decision

Why it matters: prevents training/routing rules on rejected label batches.

3.2 Golden research log + KPI board

Primary commit: [b11e2e4](#) / [2aa4b45](#)

Area	Baseline (before)	Upgrade (now)
Workflow logging	No structured end-to-end task log	lab research start action complete → research_log.db
KPIs	Doctor health only	accepted_patch_rate, tokens/success, recovery_rate, latency, portal/kpi.html
Volume	Empty or thin samples	Golden batch on starlab-demo (~26+ completions, audits n≈40+)

Why it matters: Loop 4 measure/compare needs real outcomes, not vibes.

3.3 Context packets + LEAD recovery

Primary commit: [0070c97](#)

Area	Baseline (before)	Upgrade (now)
Escalation context	Free-form paste / whole-repo stuffing	lab research packet create show — goal, files, failing tests, route, budget
Attach to task	No structured context_pack	start --packet + logstore.attach_context_pack
On failure	One-shot hope; no standard retry plan	lab research recover + lab forge run --recover (LEAD one retry + plan)

Why it matters: cleaner Grok escalations and fewer wasted long tasks after a red test.

3.4 Graph L2 — context carriage

Primary commit: [0070c97](#)

Area	Baseline (before)	Upgrade (now)
Multi-agent view	Fixed workflow / full memory dump risk	Graph as budgeted nodes: B_i under L1 envelope
Context decision	Append packing knobs only	3 gates (role, stage, recency) → pass summarize drop
CLI / telemetry	No per-node log	lab graph route-context log-node stats ablate
Ablation	None	full vs budgeted vs role_aware; save_vs_full reported

Why it matters: optimizes context carriage (RCR-style), not just model mode choice.

3.5 Shadow dual-log + graduation gates

Primary commit: [75e139d](#)

Area	Baseline (before)	Upgrade (now)
Online policy proof	Route once; no dual log	Every lab tokens route logs shadow (log-only)
Router golden	No frozen suite	modules/tokens/goldens/router_v1.jsonl + lab tokens eval

Area	Baseline (before)	Upgrade (now)
Drift	Silent heuristic edits	lab tokens eval --drift vs frozen baseline
Graduation bar	Could “learn” anytime	lab tokens gates: shadow n, join rate, golden agreement, SQC
Session continuity	i.i.d. mode switches mid tool-loop	lab tokens session lock pins mode

Why it matters: observation precedes influence; no bandit without data bar.

3.6 Four-stage progressive delivery

Primary commit: b3f0293

Area	Baseline (before)	Upgrade (now)
Release path	Jump promote / advisory canary	shadow → canary 1–5% → ramp 10/25/50 → full
Promotion criterion	Snapshot green	Window pack: success, p95 ≤1.15×, cost/success, zero safety
Rollback	Manual / untested	One-switch rollout rollback + drill
Multi-agent isolate	No role isolation	lab graph breaker trip reset

Why it matters: each stage answers a different eval question.

3.7 Design: agentic circuit breakers

Primary commit: 24e719a

Area	Baseline (before)	Upgrade (now)
Breaker model	Binary trip/reset → force budgeted	Designed FSM: Closed → DEGRADED → Open → Half-Open
DEGRADED	Not present	Mode cap, budget_frac, budgeted packing
Rollout link	Unconditional rollback risk	Breakers may only rollback when armed; never advance

Why it matters: contain cascading multi-agent failure without killing the graph.

3.8 Design: mind + body architecture

Primary commit: ced64cd

Area	Baseline (before)	Upgrade (now)
Product shape	Toolbelt / wrapper risk	Every agent = mind (Grok) + body (identity, property, contact, limits)
Body registry	None	Proposed modules/body + lab body; lab:grok-home + starlab-demo
Contact with reality	Prompt-only	Producers: forge_exit, research, ship → ledger without chat
Skills	Global ~/grok/skills	Skills land against a body

Why it matters: operators pay for durable edges, not rented intelligence alone.

3.9 Safetensors interchange

Primary commit: 6f148ec

Area	Baseline (before)	Upgrade (now)
Weight format	Ad hoc .pt / pickle possible	Publish safetensors only; thin MLX/PyTorch adapters

Area	Baseline (before)	Upgrade (now)
Package layout	Undefined	config + tokenizer + model.safetensors or sharded + index
CLI	None	lab weights validate convert publish-check
Security	Pickle code execution risk	Header-only validate; STRICT fails on pickle

Why it matters: portable, safer local heads for Apple Silicon.

3.10 Model manifest + schema

Primary commit: 892fcc3

Area	Baseline (before)	Upgrade (now)
Version metadata	Optional ad hoc notes	manifest.json with JSON Schema; schema_version ≠ model version
Integrity	No sealed hash	weights_sha256; seal CLI
Lineage	None	parent_version, code_commit, data_version, body_id, metrics
Publish	Manual checklist	publish-check --strict requires sealed manifest

Why it matters: reproducibility, rollback, canaries, cross-runtime loaders.

3.11 Tokenizer-major versioning

Primary commit: e571224

Area	Baseline (before)	Upgrade (now)
Tokenizer changes	Silent file edits possible	Own artifact + tokenizer_manifest; freeze by default
Breaking rule	Undefined	Vocab/merge/special change → major + new package
Model bind	Optional id/version only	Must set tokenizer_hash; bind-tokenizer; assert_compatible
Growth	Undefined forks	append_only + provenance; dynamic discouraged

Why it matters: silent tokenizer drift corrupts routing, horizon, and evals.

3.12 Embedding expansion drift

Primary commit: 9e62600

Area	Baseline (before)	Upgrade (now)
After vocab growth	No structured check	Before/after same prompts; slices legacy / new_token / mixed
Metrics	None	Cosine shift, L2 displacement, NN Jaccard, task deltas
Gate	None	expected warning harmful insufficient_data
CLI	None	lab weights drift example eval

Why it matters: separate expected new-token adaptation from disturbed legacy geometry.

4. End-to-end pipeline: baseline vs upgraded

Area	Baseline (before)	Upgrade (now)
Task start	Optional tokens route; free-form work	tokens route (+ shadow) → optional packet → research start
Multi-step / subagents	Deep mode packing only	graph route-context per role under budget; breakers isolate
Validation fail	Manual retry	research recover / forge --recover; recovery KPI
Policy change	Edit heuristics; ship	eval golden → gates → rollout shadow→canary→ramp→full
Local model publish	Save .pt somewhere	safetensors + sealed manifest + frozen tokenizer + drift eval
Success definition	Tests green / vibes	cost per successful task + KPI board

5. What did not change

- Grok Build remains the frontier mind; lab optimizes invocation, not training Grok itself.
- EV formula still reward – λ ·kilotokens; λ is a control signal.
- Free/offline-first; no required paid deps for core CLI.
- GitHub remote / OTEL still optional / credential-blocked where noted.
- Heavy MLX/ForgeRNN training still deferred until clean joins + gates.

6. Document map (upgrade → doc)

Topic	Path
Token policy	docs/TOKEN-AWARE-CONTROL-PLANE.md
Closed-loop / 4-stage gates	docs/research/CLOSED-LOOP-GATES.md
Graph context routing	docs/research/GRAPH-CONTEXT-ROUTING.md
Circuit breakers design	docs/design/2026-08-01-agentic-circuit-breakers.md
Mind + body design	docs/design/2026-08-01-mind-and-body.md
Safetensors	docs/SAFETENSORS-STANDARD.md
Model manifest	docs/MODEL-MANIFEST.md
Tokenizer versioning	docs/TOKENIZER-VERSIONING.md
Embedding drift	docs/EMBEDDING-DRIFT.md
Paper → lab map	docs/PAPER-T0-LAB-MAP.md

7. Suggested next implementation order

1. Implement mind/body PR-B1–B2 (registry + forge_exit ledger proof).
2. Implement agentic breaker FSM PR1 (Closed/DEGRADED/Open/Half-Open).
3. Grow shadow volume + join rate until lab tokens gates allows canary without --force.
4. Use lab weights publish-check --strict + drift eval before any local head promote.

Generated for the operator of this Grok home machine. Baseline = original integrated Star Lab; each upgrade is compared to that baseline.
PDF generated from main documentation and commit history.